Forecasting Foreign Exchange Reserves For Sri Lanka : A multivariate time series approach

Abstract

This paper compares classical econometric, Bayesian, regime-switching, and machine-learning models for forecasting Sri Lanka's foreign exchange reserves over 2005–2025, including the 2022 sovereign default episode. Using rolling-origin evaluation across five variable sets, we find that Markov-Switching VAR in a disaggregated balance-of-payments specification delivers the strongest point-forecast performance, reducing RMSE by roughly three-quarters relative to a random-walk benchmark in the post-default window. A factorial decomposition shows that model architecture explains more of this gain than simply expanding the information set, and that balance-of-payments disaggregation is most effective when paired with regime-switching structure. These results support regime-aware, component-level reserve forecasting as a practical framework for policy surveillance in crisis-prone emerging markets.

Keywords: Foreign exchange reserves, forecasting, Markov-switching VAR, balance of payments disaggregation, regime dynamics, machine learning, Sri Lanka, sovereign default, emerging markets

JEL Classification: C52, C53, E47, F31, F37

1 Introduction

Foreign exchange reserves constitute the first line of defence for emerging market economies against balance-of-payments crises, currency runs, and sovereign debt distress. Sri Lanka's experience between 2019 and 2022 illustrates the point starkly. Gross reserves declined from approximately US$7.6 billion at end-2019 to an estimated US$50 million of usable reserves by April 2022, precipitating the country's first sovereign default since independence in 1948 [Athukorala, 2024, Wignaraja, 2024]. The fallout was severe: GDP contracted by more than 7%, inflation surpassed 50%, and widespread social upheaval toppled the sitting government [Weerakoon and Jayasuriya, 2023].

Yet the academic literature on reserve forecasting remains remarkably thin. Compared to the extensive body of work on exchange rate prediction, inflation forecasting,

and GDP nowcasting, the reserve forecasting shelf is nearly bare. The few existing studies tend to rely on single-model frameworks—typically ARIMA or reduced-form VARs—and rarely evaluate performance across the kind of regime change that makes forecasting most consequential. This paper addresses that gap. It conducts a multi-model forecasting comparison applied to emerging market reserve dynamics, evaluated against a dataset spanning stable accumulation, pandemic disruption, sovereign default, and IMF-supervised recovery.

The forecasting framework draws from four methodological traditions. Classical econometrics contributes ARIMA with exogenous regressors, VECM, and their Markov-switching extensions (MS-VAR, MS-VECM). Bayesian methods contribute a BVAR with Minnesota prior, estimated via Gibbs sampling. Dynamic model combination is represented by DMA and DMS, which adaptively reweight the constituent forecasts over time. Machine learning contributes XGBoost with extensive feature engineering, XGB-Quantile for probabilistic prediction, and LSTM networks for sequence modelling. Naïve benchmarks—random walk and seasonal naïve—anchor the comparison in the tradition established by Meese and Rogoff [1983]. Additional specifications include a balance-of-payments identity model and a local-level stochastic volatility model.

Each model class is evaluated across five variable sets of increasing dimensionality: a Parsimonious set (reserves, trade balance, exchange rate; $k = 3$), a Balance of Payments set (reserves, exports, imports, remittances, tourism; $k = 5$), a Monetary set (reserves, exchange rate, M2; $k = 3$), a PCA-derived set (reserves plus three principal components extracted from eight indicators; $k = 4$), and a Full set (all available predictors; $k = 9$). This factorial design—multiple models crossed with multiple information sets—allows the effect of model architecture to be formally disentangled from the effect of information content through a difference-in-differences decomposition, a distinction that is critical when small samples and structural breaks can cause richer models to overfit rather than improve.

The evaluation framework is built around a temporally motivated train-validation-test split that isolates three distinct macroeconomic regimes: a pre-crisis training period through December 2019 (180 observations), a crisis-era validation window spanning COVID-19 and sovereign default (January 2020–December 2022), and a post-default test period covering the IMF programme and early recovery (January 2023–December 2025). Rolling-window backtests with expanding estimation windows supplement the single-split results. Because variable coverage differs across information sets, effective windows vary by variable set; we therefore report effective sample sizes explicitly. Model comparisons are formalised through Diebold and Mariano [1995] tests and the Model Confidence Set procedure of Hansen et al. [2011].

Four principal findings emerge. First, Markov-Switching VAR dominates across both the parsimonious and Balance of Payments specifications, achieving RMSE of 312–315

during the post-crisis period—a 74–77% improvement over the naïve benchmark. When Dynamic Model Averaging and Dynamic Model Selection are incorporated into the unified evaluation, the 90% Model Confidence Set retains four models: DMA, DMS, MS-VAR, and MS-VECM. Pairwise Diebold-Mariano evidence is strong—30 of 36 comparisons are significant at the 10% level (28 at 1%)—though the short, overlapping evaluation window means these p-values should be interpreted with caution. Second, a 2×2 difference-in-differences decomposition—crossing model class (MS-VAR vs. XGBoost) with information set (parsimonious vs. BoP)—reveals that the architecture effect is large: MS-VAR beats XGBoost by approximately 450 RMSE points on average, with a negative interaction term under the sign convention in Section 4.4 (DiD ≈ -242.1), indicating that moving to BoP disproportionately degrades XGBoost relative to MS-VAR. Third, a formal information-loss analysis demonstrates that aggregating vector BoP flows $\mathbf{X}_t$ into a scalar net flow A_t discards approximately 94% of the available signal (mean cancellation index ≈ 0.063), consistent with severe cancellation between offsetting gross flows during crisis episodes. Fourth, the Meese-Rogoff puzzle does not hold in its strong form for reserves: MS-VAR substantially outperforms the naïve benchmark even during the acute crisis segment, indicating that regime-switching models can beat the random walk when applied to appropriately disaggregated data.

Beyond the model horse-race, the paper contributes mechanistic evidence for *why* regime-switching models succeed. Impulse response analysis reveals strong regime asymmetry, with peak reserve responses to own shocks differing by a factor of roughly three across regimes (mean absolute peak delta ≈ 97.3). Regime characterisation shows highly persistent transition dynamics (self-transition probabilities of 0.943 and 0.983), expected durations of 17.6 and 57.2 months, and near-perfect classification certainty (mean maximum state probability 0.997). These features are economically consistent with distinct crisis and recovery phases where reserve accumulation and depletion dynamics change qualitatively by regime and by flow composition.

The paper makes three categories of contribution. *Empirically*, it provides the first systematic comparison of classical, Bayesian, regime-switching, and machine learning forecasting approaches applied to emerging market reserve dynamics, evaluated on a dataset spanning stable accumulation, sovereign default, and IMF-supervised recovery. *Methodologically*, it introduces a factorial decomposition that formally disentangles architecture from information content, demonstrates that regime-switching models applied to disaggregated balance-of-payments components substantially outperform all alternatives including strong machine learning baselines, and provides a formal test for information loss under flow aggregation. *For policy*, it contributes a component-level scenario analysis framework directly applicable to central bank reserve management and IMF Article IV surveillance, with quantified sensitivity to shocks including currency depreciation, export disruption, and programme disbursement delays.

2 Background and Literature Review

2.1 Reserve Adequacy and Early Warning Systems

The reserve adequacy literature has produced a succession of static threshold metrics—from import-cover rules and the Guidotti-Greenspan ratio [Greenspan, 1999] to monetary-based benchmarks calibrated against broad money [Calvo, 1996, Wijnholds and Kapteyn, 2001]—which the IMF's composite Assessing Reserve Adequacy (ARA) framework attempted to synthesise [IMF, 2011, 2015]. Obstfeld et al. [2010] shifted attention to financial-sector liabilities as the relevant scale variable, whilst Jeanne and Rancière [2011] formalised the cost-benefit calculus of reserve holdings as a precautionary savings problem. Both approaches imply that adequacy is fundamentally forward-looking, yet neither produces forecasts of the reserve trajectory itself. A parallel early warning systems literature [Kaminsky et al., 1998, Frankel and Saravelos, 2012] has confirmed that the pre-crisis level of reserves is the single most robust predictor of crisis incidence, but these systems predict binary crisis events rather than continuous reserve paths. The present study addresses this gap by forecasting reserve trajectories across a sample that includes an actual sovereign default, providing the dynamic complement to static adequacy assessments.

2.2 The Nature and Determinants of Reserves

Reserves are not simply a residual of balance-of-payments accounting. They are a policy variable, actively managed by central banks to achieve multiple—and sometimes conflicting—objectives. Aizenman and Lee [2007] formalised the distinction between *precautionary* motives, where buffers are built against sudden stops, and *mercantilist* motives, where reserve accumulation supports an undervalued exchange rate to promote export competitiveness. The two motives generate different accumulation dynamics. They also generate different vulnerabilities to depletion.

This dual nature matters for forecasting. Under precautionary logic, reserves co-move with external liabilities, capital account openness, and measures of financial fragility [Obstfeld et al., 2010]. Under mercantilist logic, they track trade surpluses and sterilisation costs. Calvo et al. [2013] showed that optimal reserve holdings depend critically on the probability and expected severity of sudden stops—implying that any forecasting model must account for tail-risk dynamics rather than average-case behaviour alone. Ghosh et al. [2017] documented that emerging market reserve accumulation accelerated sharply after the Asian crisis, creating a structural break in the level and trend of global reserves that complicates long-horizon extrapolation.

A related strand examines reserves as a signalling device. Adequate reserves deter speculative attacks; inadequate reserves invite them. This creates a threshold nonlinearity: the marginal value of an additional dollar of reserves is negligible when buffers are

large, but becomes very high near the critical zone where market confidence collapses [Jeanne and Rancière, 2011]. For forecasters, the implication is clear. Linear models are fundamentally misspecified near crisis thresholds, because the relationship between reserves and their determinants changes qualitatively depending on the level of reserves itself.

2.3 Previous Reserve Forecasting Literature

Despite the policy importance of reserve trajectories, the dedicated forecasting literature is remarkably thin. Early contributions applied univariate time-series methods. Sehgal and Sharma [2008] used ARIMA models to forecast Indian reserves, finding reasonable in-sample fit but limited out-of-sample reliability. Dooley et al. [1989] approached the problem indirectly through capital-flight estimation, modelling reserve changes as the residual of balance-of-payments identity relationships. This shifts the forecasting burden to the component flows without directly addressing nonlinearities in the aggregate.

More recent work has applied richer specifications. Gosselin and Parent [2005] estimated a panel error-correction model for reserve demand across 12 Asian and Latin American economies, documenting significant long-run relationships between reserves, imports, and short-term debt. Their model performed well during tranquil periods. It broke down in crisis episodes—precisely when accurate forecasts are most valuable. Aizenman [2012] extended the demand-for-reserves framework to incorporate the "fear of losing reserves" motive, showing that the speed of reserve depletion during the 2008–09 crisis was itself a function of pre-crisis adequacy. This finding implies a feedback loop that single-equation models cannot capture.

Machine learning has been applied to related problems, though evidence remains scattered. Yu et al. [2007] used support vector regression to forecast Chinese reserves with some success at short horizons. Neural network approaches have been explored for India and China, but typically lack the systematic out-of-sample evaluation protocols that are standard in the broader macroeconomic forecasting literature [Rossi, 2013]. No existing study benchmarks reserve forecasts across multiple model families—classical, Bayesian, regime-switching, and machine learning—on identical data with formal statistical comparison.

Two features of this sparse literature stand out. First, nearly all existing studies evaluate performance during stable accumulation periods. None systematically assesses accuracy *through* a sovereign default. Second, the literature overwhelmingly treats reserves as a scalar aggregate, ignoring the compositional dynamics of the underlying balance-of-payments flows. Both gaps are consequential: the former means that forecast accuracy is tested only when it matters least, and the latter means that severe information loss from flow cancellation goes undetected.

2.4 Forecasting Approaches for Macroeconomic Systems

The foundational challenge for macroeconomic forecasting was established by Meese and Rogoff [1983]. Structural exchange rate models based on monetary fundamentals could not outperform a simple random walk out of sample. The finding proved remarkably robust, replicated across dozens of currencies and time periods [Rossi, 2013, Cheung et al., 2005]. It became one of the most durable negative results in empirical macroeconomics.

The logic extends naturally to reserves. Reserves are driven by the same balance-of-payments fundamentals—trade flows, capital movements, debt service, central bank intervention—and exhibit similar nonlinearities and regime changes. Taylor et al. [2001] argued that nonlinear mean reversion becomes detectable when deviations from fundamentals are large, suggesting that the Meese-Rogoff puzzle may break down in periods of strong directional trends. The results in this paper confirm that conjecture more forcefully than anticipated.

For multivariate systems, the Vector Autoregression introduced by Sims [1980] offers a flexible, atheoretical framework treating all variables as endogenous. When variables share long-run equilibrium relationships, the Vector Error Correction Model of Johansen [1988, 1991] embeds cointegration constraints that anchor short-run dynamics to economically meaningful attractors. In the reserves context, the cointegrating relationship between reserves, trade flows, and the exchange rate provides a natural error-correction mechanism.

The Markov-switching extension is particularly relevant here. Introduced by Hamilton [1989] and generalised to multivariate systems by Krolzig [1997], it allows all model parameters to shift between discrete states governed by an unobserved Markov chain. Reserve dynamics exhibit qualitatively different behaviour during accumulation, crisis depletion, and recovery—exactly the kind of state dependence this architecture is designed to capture. Peria [2002] and Brunetti et al. [2007] applied Markov-switching models to speculative attacks in European and Southeast Asian settings, establishing precedent for regime-switching analysis in emerging market crises. Neither study, however, applied the framework to reserve forecasting with disaggregated balance-of-payments flows.

Bayesian methods address the curse of dimensionality in VAR models through informative priors. The Minnesota prior, developed by Litterman [1986] and Doan et al. [1984], centres on a random walk representation with diminishing influence from distant lags and other variables. Banbura et al. [2010] showed that BVARs with appropriately calibrated Minnesota priors match factor models even with dozens of variables. Carriero et al. [2015] demonstrated that simple fixed-hyperparameter specifications often perform as well as more elaborate alternatives. For emerging market applications with scarce data and frequent structural breaks, the prior acts as a regulariser—directly relevant for Sri Lanka, where the most data-demanding variable sets begin only in 2012, yielding roughly

95 training observations.

A more recent Bayesian strand addresses model uncertainty directly. Dynamic Model Averaging (DMA) and Dynamic Model Selection (DMS), introduced by Raftery et al. [2010] and applied to macroeconomic forecasting by Koop and Korobilis [2012], allow the weights assigned to competing models to evolve over time. DMA constructs a time-varying convex combination of model forecasts. DMS selects the single highest-probability model at each forecast origin. The appeal for reserve forecasting is clear: when regime changes alter which model best describes the data-generating process, static model selection can lock in a specification that was optimal in one state but misleading in another. DMA and DMS adapt automatically, provided the constituent models span the relevant parts of the model space.

Gradient-boosted decision trees, particularly XGBoost [Chen and Guestrin, 2016], have emerged as strong competitors to traditional econometric approaches. Medeiros et al. [2021] found tree-based methods competitive with or superior to ARIMA, factor models, and penalised regressions for Brazilian inflation. LSTM networks [Hochreiter and Schmidhuber, 1997], designed for long-range temporal dependencies, have been applied to financial and macroeconomic series with mixed results. Their performance is sensitive to hyperparameter tuning, sequence length, and training set size—constraints that bind tightly when monthly data yield at most a few hundred observations. The broader evidence suggests machine learning offers its greatest advantage in stable periods with strong nonlinear patterns. During structural breaks, simpler approaches often dominate [Richardson et al., 2021].

Rigorous forecast evaluation requires normalised accuracy metrics and formal statistical tests. The Mean Absolute Scaled Error [Hyndman and Koehler, 2006] normalises by in-sample naïve forecast error. The Diebold-Mariano test [Diebold and Mariano, 1995] tests equal predictive accuracy between competing forecasts. The Model Confidence Set [Hansen et al., 2011] identifies the subset of models containing the best performer with a given probability—valuable when many pairwise comparisons raise the risk of spurious findings.

Despite this rich methodological toolkit, four gaps persist. First, no study directly compares classical, Bayesian, regime-switching, and machine learning approaches for reserve forecasting on identical data, variable sets, and evaluation criteria. Second, regime-switching models have been applied to exchange rates and speculative attacks but never to disaggregated balance-of-payments components driving reserve dynamics. Third, evaluation is almost exclusively conducted over stable periods; systematic assessment of forecast performance *through* a sovereign default is absent. Fourth, the role of dynamic model combination methods—DMA and DMS—in reserve forecasting remains entirely unexplored. This paper addresses all four gaps simultaneously, and further contributes a formal factorial decomposition of architecture versus information effects absent from the

existing comparison literature.

2.5 The Sri Lankan Reserve Crisis

Sri Lanka's gross reserves declined from approximately US$7.6 billion at end-2019 to an estimated US$50 million of usable reserves by April 2022, when the government suspended external debt service for the first time since independence [Athukorala, 2024, Wignaraja, 2024]. The collapse was driven by the confluence of tourism revenue loss (COVID-19 and the 2019 Easter attacks), fiscal deterioration, and a Central Bank policy of defending the pegged exchange rate through reserve sales—draining gross reserves from US$3.1 billion to under US$2 billion between September 2021 and March 2022 alone. An IMF Extended Fund Facility (US$2.9 billion) was approved in March 2023, and by early 2024 gross reserves had rebuilt to approximately US$4.4 billion [Weerakoon and Jayasuriya, 2023, UNDP, 2022]. This chronology generates a reserve series with at least three distinct regimes—gradual accumulation (2005–2019), rapid depletion (2020–2022), and post-default rebuilding (2023–present)—each with fundamentally different dynamics, volatility, and relationships to driving variables, providing a particularly stringent test of model robustness and motivating the emphasis on regime-switching specifications.

3 Data and Variable Construction

3.1 Data Sources and Sample

The dataset comprises nine monthly time series spanning January 2005 to December 2025 (252 observations), drawn from the Central Bank of Sri Lanka (CBSL) statistical database, the IMF's International Reserves and Foreign Currency Liquidity (IRFCL) template, the Colombo Stock Exchange market statistics, and Sri Lanka Customs trade returns. All monetary variables are denominated in US dollars to ensure comparability and to avoid conflating real dynamics with exchange rate valuation effects. Table 1 reports descriptive statistics for all series.

3.2 Dependent Variable

Gross official reserves (USD millions) constitute the primary dependent variable. Gross reserves serve as the numerator of every major adequacy benchmark, from the import-cover rule to the Greenspan–Guidotti ratio and the IMF's composite ARA metric [Jeanne and Rancière, 2011, Obstfeld et al., 2010]. Joint ADF and KPSS testing confirms that gross reserves are integrated of order one; the first-differenced series (reserve change) is therefore the primary forecast target for the regime-switching systems. For MS-VAR and MS-VECM, predicted changes are mapped back to levels by cumulative summation

Table 1: Descriptive Statistics of Key Variables

Variable	Obs	Mean	Std Dev	Min	Max
Gross Reserves (USD m)	252	5450.6	2105.4	1588.4	9935.8
Exports (USD m)	224	886.8	176.8	282.3	1302.2
Imports (USD m)	225	1487.3	315.8	606.3	2241.0
Trade Balance (USD m)	224	−601.1	227.8	−1100.6	−39.1
Remittances (USD m)	202	507.8	121.8	204.9	812.7
Tourism (USD m)	192	171.4	128.0	0.0	475.2
Exchange Rate (LKR/USD)	228	177.0	77.3	107.6	363.3
M2 (USD m)	225	25436.4	12829.8	7095.5	48091.1
CSE Net Flows (USD m)	166	3.6	26.3	−62.2	143.8

Sample period: 2005-01 to 2025-12 (252 months). Exchange rate expressed as LKR per USD.

from the last observed reserve level. The naïve benchmark is a random walk in levels, $\hat{y}_{t+h|t} = y_t$ for all h (equivalently, zero expected change). All reported point metrics (RMSE, MAE, MAPE, MASE) are computed on level forecasts against observed levels for every model, including the benchmark, so benchmark and structural models are evaluated on an identical target scale. The first-differenced series exhibits a negative first-order autocorrelation (ACF(1) $= -0.301$), indicating mean-reverting intervention behaviour that differs materially between accumulation and crisis regimes—precisely the state-dependent behaviour the Markov-switching framework is designed to capture.

3.3 Variable Set Design

Five variable sets of increasing dimensionality are defined to disentangle the effect of model specification from the effect of information content. This design is motivated by the well-documented tension in small-sample macroeconomic forecasting between information gains from additional predictors and estimation error from parameter proliferation [Banbura et al., 2010, Carriero et al., 2015]. By evaluating each model across all five sets, the analysis identifies whether forecast failures stem from model misspecification or from over- or under-parameterisation.

The **Parsimonious set** ($k = 3$: reserves, trade balance, exchange rate) captures the first-order determinants of reserve dynamics under the import-cover and exchange rate defence frameworks. The trade balance, computed as FOB exports minus CIF imports, serves as the aggregate current account proxy. The USD/LKR exchange rate is fundamentally endogenous to reserve dynamics under Sri Lanka's managed float, since central bank intervention to defend the peg directly depletes reserves [Obstfeld et al., 2010].

The **Balance of Payments set** ($k = 5$) disaggregates the current account into ex-

Table 2: Variable Set Specifications and Effective Samples

Variable Set	Key Variables	k	Effective Coverage	Effective N
Parsimonious	Reserves, trade balance, exchange rate	3	2007:01–2025:12	156
BoP	Reserves, exports, imports, remittances, tourism	5	2009:01–2025:03	132
Monetary	Reserves, exchange rate, M2	3	2005:01–2025:06	180
PCA	Reserves + 3 PCs from 8 indicators	4	2012:01–2024:11	95
Full	All available predictors	9	2012:01–2025:03	96

Notes: k = number of endogenous variables in the VAR system. Effective N (train) counts observations up to 2019:12 after applying the common preprocessing pipeline for each variable set. Missing-data handling follows a uniform rule before evaluation (forward-fill up to 3 consecutive months, then drop remaining missing rows); no model-specific imputation is used in the horse-race stage. The BoP set replaces trade balance with separate exports and imports to identify independent shocks. The PCA set extracts three principal components from eight indicators, fitted on training data only to avoid look-ahead bias.

ports, imports, remittances, and tourism earnings alongside reserves. The disaggregation is driven by the requirements of the MS-VAR framework, which needs to identify independent shocks to each flow component. This is critical because BoP components exhibit sharply divergent crisis dynamics: during the 2020–2022 period, exports and imports moved in opposite directions, whilst remittances and tourism followed distinct structural break patterns.

The **Monetary set** ($k = 3$) comprises reserves, the exchange rate, and broad money (M2), introducing the domestic capital flight channel identified by Obstfeld et al. [2010] and captured in the IMF's ARA metric. This compact specification tests whether monetary aggregates provide incremental forecasting power beyond trade flows.

The **PCA set** ($k = 4$) comprises reserves plus three principal components extracted from the eight indicators in the broader dataset, testing whether dimensionality reduction can preserve informational content whilst reducing estimation burden—particularly relevant for the BVAR and MS-VAR specifications, where parameter proliferation in small samples is a binding constraint. The three retained components explain 79.5% of total variance (PC1: 46.1%, PC2: 20.8%, PC3: 12.7%).

The **Full set** ($k = 9$) includes all available predictors simultaneously, serving as an upper bound on available information and as a test of whether regularisation mechanisms—Minnesota shrinkage, tree-based feature selection, LSTM dropout—can effectively manage the dimensionality.

Table 2 highlights the ragged-edge implications of uneven source coverage: variable sets are estimated on different effective windows, but all models within a given variable set are evaluated on the same preprocessed sample. Cross-model comparisons within a variable set are therefore sample-aligned; cross-variable-set comparisons should be interpreted with the effective-sample caveat in mind.

3.4 Stationarity and Seasonal Properties

All series are subjected to joint ADF and KPSS testing to determine the order of integration. Variables confirmed as I(1) are first-differenced for VAR estimation; where cointegrating relationships are identified via the Johansen procedure, the VECM specification is estimated in levels with error-correction terms. Seasonal properties are assessed via STL decomposition. Variables exhibiting seasonal strength above 0.5—notably tourism earnings (0.593)—are deseasonalised using X-13ARIMA-SEATS prior to estimation in specifications that do not include seasonal dummies.

4 Methodology

4.1 Model Overview

The model set spans classical econometrics (ARIMA, VAR, VECM), regime-switching specifications (MS-VAR, MS-VECM), Bayesian approaches (BVAR), dynamic model combination (DMA, DMS), machine learning (XGBoost, LSTM, XGB-Quantile), and naïve benchmarks (random walk, seasonal naïve). Additional specifications include a balance-of-payments identity model (BoPIdentity) and a local-level stochastic volatility model (LocalLevelSV). All models are estimated in an expanding-window fashion, initialising on the training period (January 2005–December 2019, 180 observations) and expanding monthly through the validation (2020–2022) and test (2023–2025) periods. Standard estimation details are reported in Appendix A.

This section focuses on the non-standard modelling choices that distinguish the present application from textbook implementations.

4.2 Regime-Switching Specifications

The MS-VAR allows all parameters to shift between two discrete regimes (accumulation and depletion) governed by an unobserved Markov chain with state $S_t \in \{0, 1\}$:

$$\mathbf{y}_t = \mathbf{c}_{S_t} + \sum_{i=1}^{p} \mathbf{A}_{i,S_t} \mathbf{y}_{t-i} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_{S_t})$$

Parameters are estimated via the EM algorithm. The key modelling choice is the two-regime specification, motivated by the distinct accumulation and depletion dynamics visible in the reserve series. Higher-order specifications ($K = 3$) were explored but yielded poorly identified regimes with short durations, consistent with the limited within-regime observations available in a 252-month sample. At the default iteration cap, some full-sample runs can stop at the maximum iteration bound; accordingly, the regime diagnostics are complemented with a seeded multi-start stability check under a higher iteration

cap (Section 6.2).

For the MS-VECM extension, a critical design decision concerns the cointegrating vectors:

$$\Delta \mathbf{y}_t = \boldsymbol{\alpha}_{S_t} \boldsymbol{\beta}' \mathbf{y}_{t-1} + \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_{i,S_t} \Delta \mathbf{y}_{t-i} + \mathbf{u}_t$$

The cointegrating vectors $\boldsymbol{\beta}$ are estimated on the full sample and held regime-invariant, whilst the adjustment speeds $\boldsymbol{\alpha}_{S_t}$ and short-run dynamics $\boldsymbol{\Gamma}_{i,S_t}$ are regime-dependent. This follows the approach of Krolzig [1997]: the long-run equilibrium between reserves, trade flows, and the exchange rate is assumed to hold across regimes, but the speed at which the system corrects deviations from equilibrium differs between accumulation and crisis states.

4.3 XGBoost Rolling Feature Engineering

The XGBoost implementation departs from standard applications through its rolling feature engineering pipeline, which constructs 3-month and 6-month rolling means and volatility estimates for all variables at each forecast origin. This creates a feature set that evolves with the expanding estimation window, capturing momentum and volatility clustering that are economically meaningful for reserve dynamics. The rolling construction ensures that no future information leaks into the feature set. XGB-Quantile extends the framework to produce probabilistic forecasts by predicting the 10th, 25th, 50th, 75th, and 90th percentiles of the conditional distribution. Time-series cross-validation tuning is employed for XGBoost in the parsimonious and BoP specifications, though robustness is mixed across folds.

4.4 Architecture-vs-Information Decomposition

A key methodological contribution is the formal disentangling of architecture (model class) from information content (variable set). We construct a 2×2 factorial design crossing model class $\mathcal{M} \in \{\text{MS-VAR}, \text{XGBoost}\}$ with information set $\mathcal{I} \in \{\text{Parsimonious}, \text{BoP}\}$. For RMSE outcomes $R_{m,i}$, the architecture effect (holding information constant) is

$$\hat{\alpha} = \frac{1}{2} \sum_{i \in \mathcal{I}} \left[R_{\text{XGB},i} - R_{\text{MSVAR},i} \right],$$

the information effect (holding model constant) is

$$\hat{\iota} = \frac{1}{2} \sum_{m \in \mathcal{M}} \left[R_{m,\text{Parsim}} - R_{m,\text{BoP}} \right],$$

and the interaction (difference-in-differences) is

$$\hat{\delta} = (R_{\text{XGB,Parsim}} - R_{\text{XGB,BoP}}) - (R_{\text{MSVAR,Parsim}} - R_{\text{MSVAR,BoP}}) .$$

This decomposition quantifies whether the superior performance of MS-VAR on BoP data is driven primarily by the regime-switching architecture, by the richer information in disaggregated flows, or by an interaction between the two.

4.5 Dynamic Model Averaging and Selection

Dynamic Model Averaging and Dynamic Model Selection [Raftery et al., 2010, Koop and Korobilis, 2012] address model uncertainty by allowing the weights on competing forecasts to evolve over time. At each forecast origin t, DMA computes a posterior model probability $\pi_{t|t-1,k}$ for each constituent model k and forms the combined forecast as the probability-weighted average. DMS selects the single model with the highest $\pi_{t|t-1,k}$. The forgetting factor α governs how quickly past performance is discounted; we set $\alpha = 0.99$, implying an effective memory of roughly 100 periods. The rolling forecast-error variance is estimated over a 24-month window.

The constituent model pool comprises all seven non-DMA/DMS models evaluated on the BoP specification: ARIMA, BVAR, MS-VAR, MS-VECM, Naïve, VECM, and XGBoost. This design means DMA and DMS can dynamically shift weight toward regime-switching models during crisis periods and toward simpler specifications during tranquil periods—an adaptive strategy that static model selection cannot replicate.

4.6 Information Loss Under Aggregation

To test whether aggregating balance-of-payments components into composite variables destroys predictive information, we formalise the aggregation mapping. Let $\mathbf{X}_t = (x_{1t}, \ldots, x_{kt})'$ denote the vector of gross BoP flows and $A_t = \mathbf{w}'\mathbf{X}_t$ the scalar aggregate (e.g., trade balance = exports − imports). Forecasting with A_t conditions on a strictly coarser σ-field than forecasting with $\mathbf{X}_t$, implying potential information loss. We quantify the empirical severity of this loss through a cancellation index:

$$\text{CI}_t = \frac{|A_t|}{\sum_{j=1}^{k} |x_{jt}|},$$

which measures the ratio of the net flow to the sum of absolute gross flows. When CI_t is near zero, large gross flows cancel almost entirely, and the aggregate retains almost no information about the underlying dynamics.

4.7 Evaluation Framework

Forecasts at horizons $h \in \{1, 3, 6, 12\}$ months are evaluated using RMSE, MAE, MAPE, and MASE [Hyndman and Koehler, 2006] for point accuracy; CRPS and prediction interval coverage probability (PICP) at the 80% and 95% levels for density forecasts; and an asymmetric policy loss function that penalises under-prediction of reserve depletion at twice the rate of over-prediction. All metrics are computed on reserve levels, and the random-walk benchmark is transformed identically to the structural models before scoring. Statistical significance is assessed via the Diebold and Mariano [1995] test of equal predictive accuracy (with Harvey-Leybourne-Newbold small-sample correction) and the Model Confidence Set of Hansen et al. [2011] with stationary block bootstrap (1,000 replications). Formal definitions of all metrics are provided in Appendix A.

5 Empirical Results

5.1 Main Forecasting Comparison

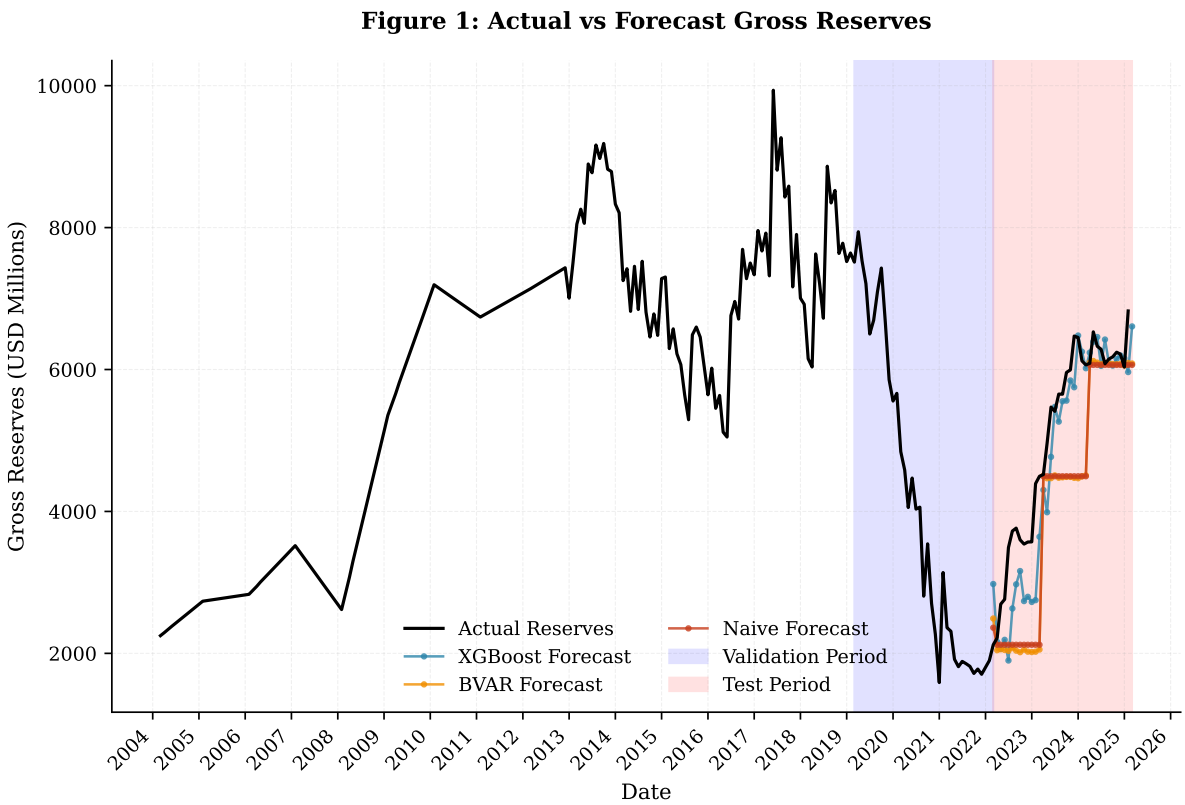

Figure 1: Actual gross reserves and one-step-ahead forecasts from selected models. The validation period (2020–2022, blue shading) covers the pandemic and sovereign default; the test period (2023–2025, red shading) covers the IMF-supervised recovery.

Table 3: Out-of-Sample Forecast Accuracy: Parsimonious Specification, $h = 1$

Model	RMSE	MAE	MAPE	MASE	Policy Loss	CRPS	n
MS-VAR	**311.8**	**230.3**	**5.20**	**0.77**	**408.4**	–	36
MS-VECM	357.7	280.5	6.31	0.94	512.7	–	36
XGBoost	640.6	475.7	11.82	1.59	900.8	466.4	36
XGB-Quantile	655.5	521.8	12.25	1.75	1018.0	241.0	36
ARIMA	1170.4	896.2	21.61	3.00	1785.6	734.2	36
Naïve	1178.9	935.9	20.47	3.14	1864.2	761.5	36
VECM	1194.5	956.8	21.64	3.21	1892.3	782.4	36
BVAR	1214.9	964.4	21.41	3.23	1917.2	792.1	36

Notes: MAPE in %. Policy Loss = asymmetric loss (2× penalty for under-prediction). CRPS = Continuous Ranked Probability Score. Bold indicates best performance. Test period: 2023:01–2025:12. Dashes indicate models without probabilistic output in this specification.

Table 3 presents out-of-sample forecast accuracy for the parsimonious specification at $h = 1$ over the test period (2023:01–2025:12). The result is striking. MS-VAR achieves RMSE of 311.8, versus 1,178.9 for the naïve random walk—a 73.6% improvement. The dominance extends across every metric: MAE of 230.3 (versus 935.9), MAPE of 5.20% (versus 20.47%), and MASE of 0.77, which is below 1.0 and therefore outperforms the naïve in-sample benchmark. MS-VECM follows closely at 357.7.

XGBoost sits in between, with RMSE of 640.6. This is substantially better than all classical and Bayesian specifications, but roughly double the MS-VAR error. Classical models (ARIMA, VECM) cluster in the range 1,170–1,195—indistinguishable from the naïve benchmark. BVAR fares no better at 1,214.9, reflecting the difficulty of estimating a high-dimensional system with limited pre-crisis training data.

The separation is clear. Regime-switching models occupy one tier (RMSE 312–358). Everything else sits in another (RMSE > 640). This gap motivates the mechanistic investigation in Section 6.

5.2 Statistical Significance

Table 4 reports Diebold-Mariano tests for key model pairs. MS-VAR's advantage over all non-regime-switching, non-combination models is highly significant ($p < 0.01$ in every case). The MS-VAR vs. XGBoost comparison yields $p = 0.0012$, confirming that the regime-switching advantage is not a statistical artefact. Across the full nine-model set in the BoP specification (including DMA and DMS), 30 of 36 pairwise DM comparisons are significant at the 10% level, and 28 at the 1% level.

The Model Confidence Set at $\alpha = 0.10$ retains four models: MS-VAR, DMA, MS-VECM, and DMS. All five remaining models—XGBoost ($p = 0.009$), ARIMA, Naïve, BVAR, and VECM (all $p < 0.001$)—are eliminated. This result is decisive but also

Table 4: Diebold-Mariano Test Results: Selected Model Pairs

Model Pair	DM p-value	Significance
MS-VAR vs. ARIMA	0.0006	***
MS-VAR vs. BVAR	<0.0001	***
MS-VAR vs. VECM	<0.0001	***
MS-VAR vs. Naïve	<0.0001	***
MS-VAR vs. XGBoost	0.0012	**
MS-VECM vs. XGBoost	0.0033	**
XGBoost vs. ARIMA	0.0052	**

Notes: Diebold-Mariano test with Harvey-Leybourne-Newbold small-sample correction. Two-sided p-values. ***: $p < 0.001$; **: $p < 0.01$. Pairwise matrix totals cited in the text refer to the nine-model BoP specification, where the effective $h = 1$ test window is 2023:01–2025:03 ($n = 27$).

informative: the MCS does not single out one winner. Instead, it draws a sharp boundary between regime-aware methods (including dynamic combinations that lean heavily on regime-switching constituents) and everything else.

These inference results should be interpreted with appropriate caution. The BoP test window contains 27 effective monthly observations at $h = 1$, and overlapping multi-step horizons further reduce effective independence (roughly $n_{\text{eff}} \approx n/h$: about 9 at $h = 3$, 4–5 at $h = 6$, and 2–3 at $h = 12$). In short samples, both DM asymptotics and block-bootstrap MCS outcomes are sensitive to tuning choices such as block length. We therefore treat p-values as complementary evidence and place primary weight on rank stability and economic magnitude: in the BoP $h = 1$ test window, MS-VAR lowers RMSE from 1,350.8 (naïve) to 315.2, a reduction of 1,035.6 USD million—about 22.3% of mean test-period reserves.

5.3 Horizon Sensitivity

Table 5: Forecast Accuracy Across Horizons: RMSE (Parsimonious Specification)

Model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	h_{12}/h_1
XGBoost	**640.6**	**1281.7**	1887.6	2664.2	4.16
Naïve	1178.9	1393.2	**1737.4**	**2373.7**	2.01
VECM	1194.5	1404.6	1751.5	2398.4	2.01
BVAR	1214.9	1438.0	1793.5	2488.9	2.05
XGB-Quantile	655.5	1384.5	1991.1	3095.0	4.72
ARIMA	1170.4	1561.6	2453.0	3444.6	2.94

Notes: h_{12}/h_1 = deterioration ratio from 1-month to 12-month horizon. Bold indicates best performance at each horizon. Test period: 2023–2025. Parsimonious MS-VAR and MS-VECM horizon results are omitted here to avoid duplication with Table 7, which reports the preferred BoP regime-switching specification used for the main regime comparisons.

Table 5 reveals pronounced horizon dependence among non-regime-switching models. XGBoost dominates at short horizons ($h = 1$: 640.6; $h = 3$: 1,281.7). But this advantage deteriorates. By $h = 6$, the naïve random walk (1,737.4) overtakes XGBoost (1,887.6). The pattern reflects a fundamental limitation of machine learning in macroeconomic forecasting: the model exploits patterns in the recent training distribution, and as the horizon extends, distributional shift erodes that advantage. Bayesian methods, by contrast, maintain more stable relative performance. BVAR's deterioration ratio is 2.05, versus 4.16 for XGBoost—a regularising benefit of the Minnesota prior.

5.4 Validation versus Test Period

Table 6: Forecast Performance: Crisis Validation vs. Post-Default Test Period

	Validation (2020–2022)			Test (2023–2025)		
Model	RMSE	MAE	MAPE	RMSE	MAE	MAPE
XGBoost	1183.5	1037.3	37.82	640.6	475.7	11.82
XGB-Quantile	1055.3	884.8	32.81	655.5	521.8	12.25
ARIMA	1986.3	1723.7	70.36	1170.4	896.2	21.61
Naïve	1328.1	1029.3	35.69	1178.9	935.9	20.47
VECM	1532.2	1289.6	46.57	1194.5	956.8	21.64
BVAR	1412.7	1128.9	39.59	1214.9	964.4	21.41

Notes: Validation period coincides with the Sri Lankan economic crisis and sovereign default (April 2022). Test period covers the post-default recovery under the IMF Extended Fund Facility. MAPE in %. This table focuses on the non-regime-switching parsimonious set; MS-VAR and MS-VECM period splits are reported in the BoP-focused regime tables to avoid repetitive presentation across specifications.

Table 6 stratifies results by the crisis validation period (2020:01–2022:12) and post-default test period (2023:01–2025:12). During the acute crisis, MAPE ranges from 33% to over 70% across these non-regime-switching models, indicating severe forecasting difficulty. In sharp contrast, during the post-crisis recovery, XGBoost achieves MAPE of 11.82%, and multiple models achieve MAPE in the 20–22% range. However, the regime-switching models—which are not shown in this parsimonious-only comparison but dominate the BoP results below—substantially outperform even during the crisis segment, as discussed in Section 6.6.

5.5 Balance of Payments Specification Results

Table 7 reveals a tight cluster of regime-aware models at the top of the BoP specification. MS-VAR achieves RMSE of 315.2 at $h = 1$—a 76.7% improvement over the naïve benchmark (1,350.8). DMA follows at 321.7, MS-VECM at 323.4, and DMS at 326.0. The gap separating these four models from the next competitor is large: XGBoost, at 886.1,

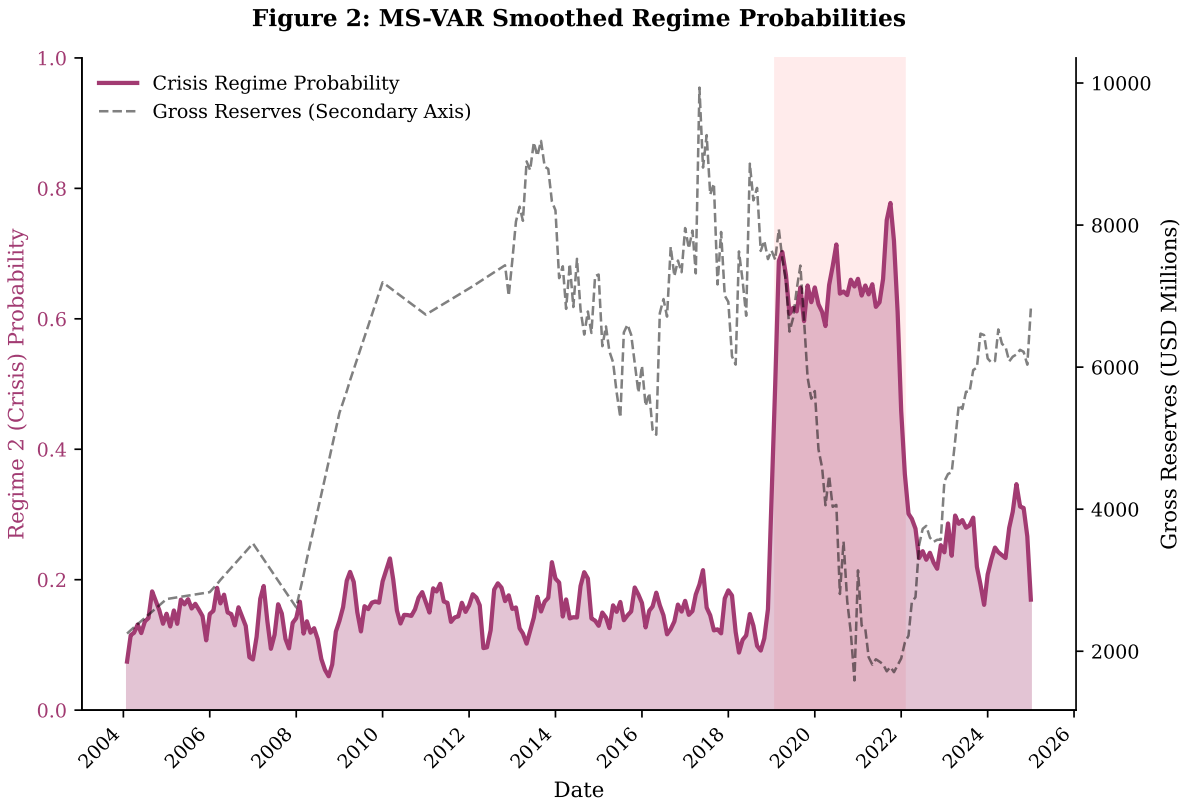

Figure 2: MS-VAR smoothed regime probabilities (top axis) and actual gross reserves (bottom axis). The estimated crisis regime probability rises sharply during 2020–2022, consistent with the pandemic disruption and sovereign default, and recedes during the post-2023 IMF-supervised recovery.

Table 7: Forecast Accuracy Across Horizons: RMSE (Balance of Payments Specification)

Model	$h = 1$	$h = 3$	$h = 6$	$h = 12$
MS-VAR	**315.2**	**679.5**	1150.7	2050.4
DMA	321.7	694.2	1143.8	2031.5
MS-VECM	323.4	842.8	1697.0	3249.8
DMS	326.0	701.8	**1049.8**	2018.3
LSTM	479.7	713.9	1057.3	**1742.2**
XGBoost	886.1	1376.1	1869.4	2679.0
BoPIdentity	1136.4	1006.3	1169.6	2734.7
ARIMA	1249.9	1463.9	1834.0	2448.6
Naïve	1350.8	1599.9	1914.3	2541.6
BVAR	1361.5	1597.3	1888.6	2487.1
LocalLevelSV	1410.0	1661.9	1964.2	2569.3
VECM	1556.3	1601.3	1921.3	2650.5

Notes: BoP set: reserves, exports, imports, remittances, tourism ($k = 5$). DMA and DMS combine forecasts from all non-combination models in this specification. Test period: 2023–2025. Bold indicates best at each horizon.

trails by more than 560 RMSE points. At longer horizons, DMS becomes increasingly competitive ($h = 6$: 1,049.8), and LSTM emerges as the strongest performer at $h = 12$ (1,742.2), suggesting that both dynamic model selection and sequence models can capture longer-range dependencies that static regime-switching specifications miss.

The pattern is revealing. DMA and DMS succeed not by inventing new information, but by dynamically reweighting the constituent models. During periods when the MS-VAR dominates, DMA assigns it the lion's share of weight. When no single model clearly leads, the combination hedge pays off. The net result is a forecast that is nearly as accurate as the best individual model at short horizons and more robust across horizons.

The success of regime-aware methods on disaggregated BoP data points to a deeper lesson. The key to accurate reserve forecasting lies in separately modelling the distinct dynamics of each balance-of-payments component—exports, imports, remittances, tourism—and allowing those relationships to shift across economic states. The parsimonious specification, which aggregates exports and imports into a single trade balance, discards information about asymmetric crisis dynamics that regime-switching models can exploit.

5.6 Variable Set Robustness

Table 8: Best Model by Variable Set: RMSE at $h = 1$ (Test Period)

Variable Set	Best Model	RMSE	vs. Naïve (%)
Parsimonious	MS-VAR	311.8	−73.6
BoP	MS-VAR	315.2	−76.7
Monetary	XGBoost	307.1	−76.3
PCA	MS-VECM	444.5	−65.7
Full	XGBoost	387.9	−70.1

Notes: The "vs. Naïve" column reports percentage RMSE reduction relative to the naïve random walk in each specification. A strong interaction pattern is evident: MS-VAR dominates in parsimonious and BoP sets, while XGBoost dominates in monetary and full sets.

Table 8 reveals a striking interaction between model class and variable set. MS-VAR dominates in the parsimonious and BoP specifications (RMSE $\approx$ 312–315), while XGBoost dominates in the monetary and full specifications (RMSE $\approx$ 307–388). This is consistent with regime models preferring compact, structurally relevant systems where state-dependent dynamics are cleanly identified, while tree models can exploit broader nonlinear feature spaces in richer specifications. The PCA specification is best served by MS-VECM (444.5), suggesting that dimensionality reduction partially preserves the regime structure that switching models exploit.

At the same time, cross-variable-set comparisons inherit different effective estimation windows (Table 2). For example, BoP estimation starts later than the monetary

specification because tourism and remittance coverage begins later. The horse race is therefore clean within each variable set, while cross-set contrasts are interpreted with this sample-alignment caveat.

6 Mechanism Analysis

6.1 Architecture versus Information: The 2×2 Decomposition

Table 9: Architecture-Information Decomposition: 2×2 RMSE Matrix

	Parsimonious	BoP	Row Mean
MS-VAR	311.8	315.2	313.5
XGBoost	640.6	886.1	763.4
Column Mean	476.2	600.7	

Notes: Architecture effect (MS-VAR advantage, averaged across varsets) ≈ 449.9 RMSE points. Information effect ≈ −124.5 (BoP is worse on average, driven by XGBoost). Interaction (DiD), using $\hat{\delta} = (R_{\text{XGB,Parsim}} - R_{\text{XGB,BoP}}) - (R_{\text{MSVAR,Parsim}} - R_{\text{MSVAR,BoP}})$, is ≈ −242.1.

Table 9 presents the factorial decomposition. The architecture effect is dominant: MS-VAR outperforms XGBoost by approximately 450 RMSE points on average across variable sets. The information effect is negative on average (−124.5), indicating that the BoP specification does not uniformly improve forecasts—it helps regime-switching models marginally but substantially hurts XGBoost, whose feature importance analysis (Section 6.5) reveals heavy reliance on autoregressive reserve features rather than rich BoP components. The interaction term (DiD ≈ −242.1 under the sign convention above) indicates that moving from Parsimonious to BoP degrades XGBoost much more than MS-VAR (245.5 RMSE points versus 3.4), consistent with BoP disaggregation being substantially more exploitable by regime-switching architecture than by tree-based models.

6.2 Regime Characterisation

Table 10 reports regime diagnostics. Both regimes are highly persistent. The crisis regime has a self-transition probability of 0.943, implying an expected duration of 17.6 months. The accumulation regime is even stickier at 0.983, with an expected duration of 57.2 months. Classification certainty is near-perfect: the mean maximum state probability is 0.997, and 99.5% of observations are classified with probability above 0.90. Entropy is correspondingly low (0.012). Classification shares (30.1% / 69.9%) differ from ergodic shares (23.5% / 76.5%), consistent with finite-sample state composition differing from long-run transition-implied frequencies.

Table 10: MS-VAR Regime Characterisation: BoP Specification

	Regime 0 (Crisis)	Regime 1 (Accumulation)
Self-transition probability	0.943	0.983
Expected duration (months)	17.6	57.2
Share of observations	30.1%	69.9%
Classification Quality		
Mean max probability	0.997	
Share with Pr $\geq$ 0.90	99.5%	
Mean entropy	0.012	
Estimation Diagnostics		
Converged	No (max iterations)	
Log-likelihood	-38.28	
Observations (differenced)	193	

Notes: Transition probabilities from EM estimation on the BoP variable set. Expected duration = $1/(1 - p_{ii})$ where p_{ii} is the self-transition probability; durations are computed from full-precision probabilities while table probabilities are rounded to three decimals. "Share of observations" reports smoothed-state classification frequencies, not ergodic long-run shares. The ergodic distribution implied by the transition matrix is approximately 23.5% (Regime 0) and 76.5% (Regime 1).

This sharp separation is economically intuitive. Reserve accumulation is gradual, driven by persistent BoP surpluses. Crisis depletion is rapid, driven by sudden stops and peg defence. The data distinguish cleanly between these states.

A caveat remains: unconstrained random initialisations can still reach lower-likelihood local optima. However, the seeded perturbation multi-start check in Table 11 shows that the economically motivated baseline solution is locally stable in both regime diagnostics and out-of-sample loss. We therefore report regime estimates together with explicit initialization and convergence diagnostics, rather than relying on a single EM run.

6.3 Impulse Response Analysis

Regime-conditional generalised impulse response functions reveal strong asymmetry in shock propagation. The reserve response to its own shock peaks at roughly 559 in the accumulation regime—three times the crisis-regime peak of approximately 173. Across all target variables, the mean absolute peak delta is 97.3. Half-life differences average 9.0 months.

The asymmetry is economically meaningful. During accumulation, reserve shocks propagate via policy responses—exchange rate management, import liberalisation—that amplify the initial impact. During crisis, feedback channels are suppressed. Binding constraints (depleted reserves, capital controls, suspended convertibility) truncate shock propagation. Linear models must average across these regimes. The MS-VAR does not,

Table 11: MS-VAR Multi-Start Stability Check (BoP Specification)

Metric	Estimation Stability (20 starts)	Forecast Stability ($h = 1$ test)
Convergence	20 / 20 runs converged	20 / 20 runs evaluated
Log-likelihood	mean −48.6; median −38.3	–
p_{00} / p_{11}	mean 0.920 / 0.934	–
Expected duration (months)	median 17.6 / 57.2	–
Classification certainty	mean max probability 0.987	–
RMSE across starts	–	mean 320.6 (SD 9.6)
RMSE range across starts	–	309.5 to 345.1
Improvement vs. Naïve (RMSE)	–	74.5% to 77.1%

Notes: Each start perturbs 5% of the baseline volatility-threshold initial state assignments and re-estimates the BoP MS-VAR with a 200-iteration EM cap. Forecast stability is computed from rolling-origin $h = 1$ RMSE on the BoP effective test window (2023:01–2025:03; $n = 27$).

and that is a key source of its forecasting advantage.

6.4 Information Loss Under Aggregation

Table 12: Information Loss from Flow Aggregation

Period	Cancellation Index	Information Loss	RMSE Improvement	Share Helped
All	0.063	93.8%	−1.68%	57.1%
Crisis (2020–2022)	0.074	92.6%	−1.68%	57.1%
Tranquil (2023–2025)	0.059	94.1%	−2.92%	57.1%

Notes: Cancellation Index = $|A_t|/\sum|x_{jt}|$ where A_t is the aggregate and x_{jt} are gross flows. Information Loss = 1 − CI. RMSE Improvement = mean percentage change when moving from aggregated to disaggregated specification. Share Helped = proportion of models for which disaggregation reduces RMSE.

Table 12 reports the empirical information loss analysis. The mean cancellation index is strikingly low at 0.063, implying that approximately 94% of the information in gross BoP flows is destroyed by aggregation into net flows. This severe cancellation arises because exports and imports are both large and move in partially offsetting directions: during the crisis, export growth of 14.3% coexisted with import contraction of 26%, producing modest net changes that mask dramatic gross flow dynamics. The RMSE improvement from disaggregation, while modest in percentage terms (−1.68% to −2.92%), is model-dependent: disaggregation helps regime-switching models (BVAR, MS-VECM) but hurts others (notably XGBoost, which shows a −20% RMSE degradation when moving to BoP). This asymmetric pattern is consistent with the architecture-information interaction identified in the 2×2 decomposition.

6.5 Feature Importance

XGBoost feature importance analysis reveals that the model relies overwhelmingly on autoregressive reserve features rather than rich BoP components. In the BoP specification, the 3-month moving average of reserves alone accounts for approximately 75% of total feature importance, with the next four features (reserve lags at horizons 1–3 and rolling standard deviation) contributing a further 13%. Disaggregated BoP variables (remittances, exports, imports, tourism) collectively account for less than 5% of importance. This finding explains why XGBoost gains little from moving to the BoP specification: it effectively ignores the additional information, treating the BoP variables as noise. By contrast, the MS-VAR framework is structurally designed to model the joint dynamics of all system variables, and its regime-switching parameters capture the state-dependent correlations between BoP components and reserves that the tree model cannot efficiently exploit.

6.6 Crisis Segment Performance and the Meese-Rogoff Puzzle

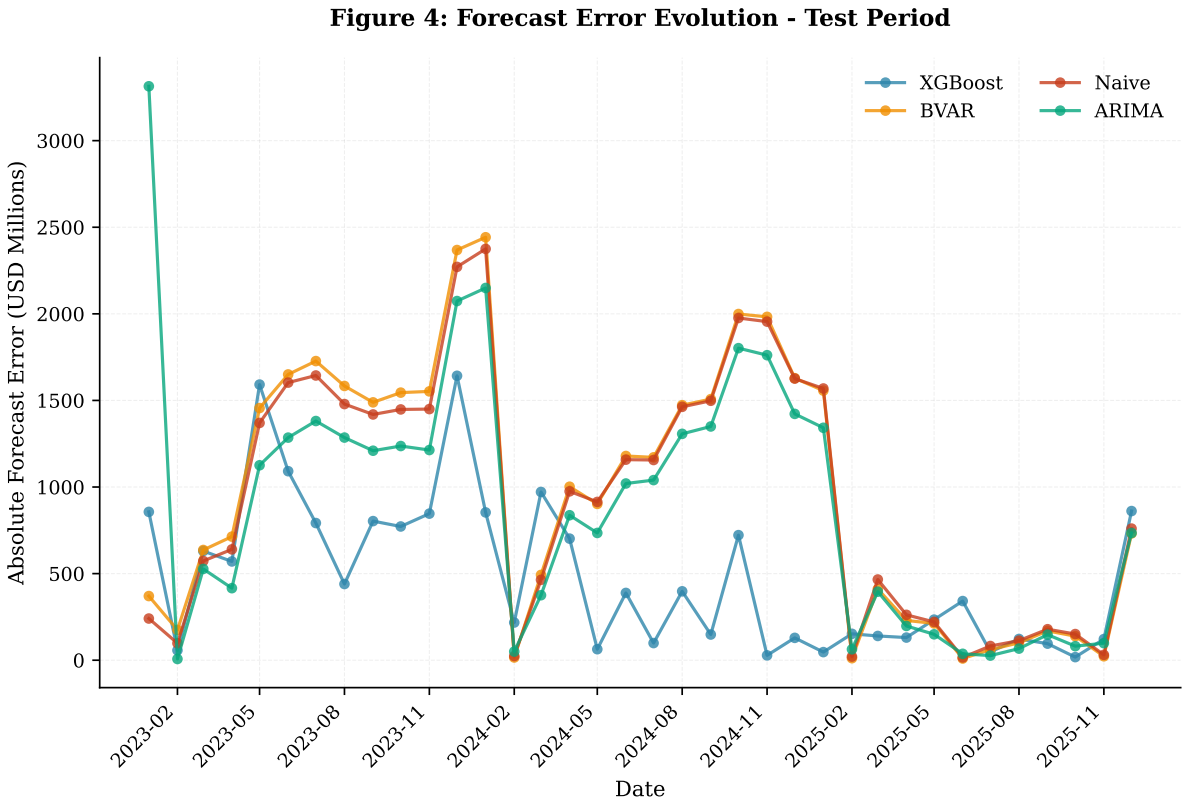

Figure 3: Absolute forecast errors over the test period (2023–2025) for selected models. MS-VAR maintains consistently lower errors, though all models exhibit episodic spikes corresponding to months of large reserve movements.

The results in Table 13 reveal that the Meese-Rogoff puzzle does not hold in its strong

Table 13: Crisis Segment Performance: Best Models by Period

Segment	Best Model	RMSE	Naïve RMSE
Crisis (2020–2022, BoP validation)	MS-VAR (BoP)	518.2	1345.5
Post-default (2023–2025, BoP test)	MS-VAR (BoP)	315.2	1350.8
All pooled (2020–2025, BoP)	MS-VAR (BoP)	438.5	1338.0

Notes: RMSE values are BoP-specification $h = 1$ results from unified rolling-origin outputs. "All pooled" combines validation and test forecasts; effective pooled window is 2020:02–2025:03 due BoP data availability.

form for reserves. The naïve random walk is beaten substantially by MS-VAR even during the 2020–2022 validation crisis segment: crisis RMSE of approximately 518 versus naïve crisis RMSE of 1,345 represents a 61.5% improvement. This finding is significant because it suggests that regime-switching models, when applied to appropriately disaggregated data, can capture the dynamics of reserve depletion during sudden stops—a result that extends beyond the standard Meese-Rogoff finding, which was formulated in the context of exchange rate models applied to aggregated fundamentals.

However, the puzzle does retain a weaker form: among non-regime-switching models, the naïve benchmark remains competitive during crisis periods, and none of ARIMA, VECM, BVAR, or XGBoost reliably beats it in the BoP crisis-validation segment. Classical and machine-learning models without explicit regime structure remain vulnerable to abrupt distributional breaks.

7 Density Forecast Evaluation

Table 14: Density Forecast Evaluation: Parsimonious Specification, $h = 1$

Model	CRPS	Log Score	Coverage_{80} (%)	Coverage_{95} (%)
XGB-Quantile	**241.0**	–	**72.2**	**97.2**
XGBoost	466.4	−928.95	2.8	5.6
ARIMA	734.2	−10.12	41.7	50.0
Naïve	761.5	−9.88	44.4	52.8
VECM	782.4	−10.30	36.1	52.8
BVAR	792.1	−10.18	38.9	50.0

Notes: Coverage_{80} and Coverage_{95} = empirical coverage rates of 80% and 95% prediction intervals. Bold indicates best. Models without probabilistic output are marked with dashes.

Table 14 evaluates probabilistic forecasting performance. XGB-Quantile is the clear winner, with CRPS of 241.0 and the best coverage rates: 72.2% at 80% nominal and 97.2% at 95% nominal. Classical specifications (ARIMA, VECM) and BVAR exhibit poor coverage (36–50% at 80% nominal), suggesting that their estimated densities sub-

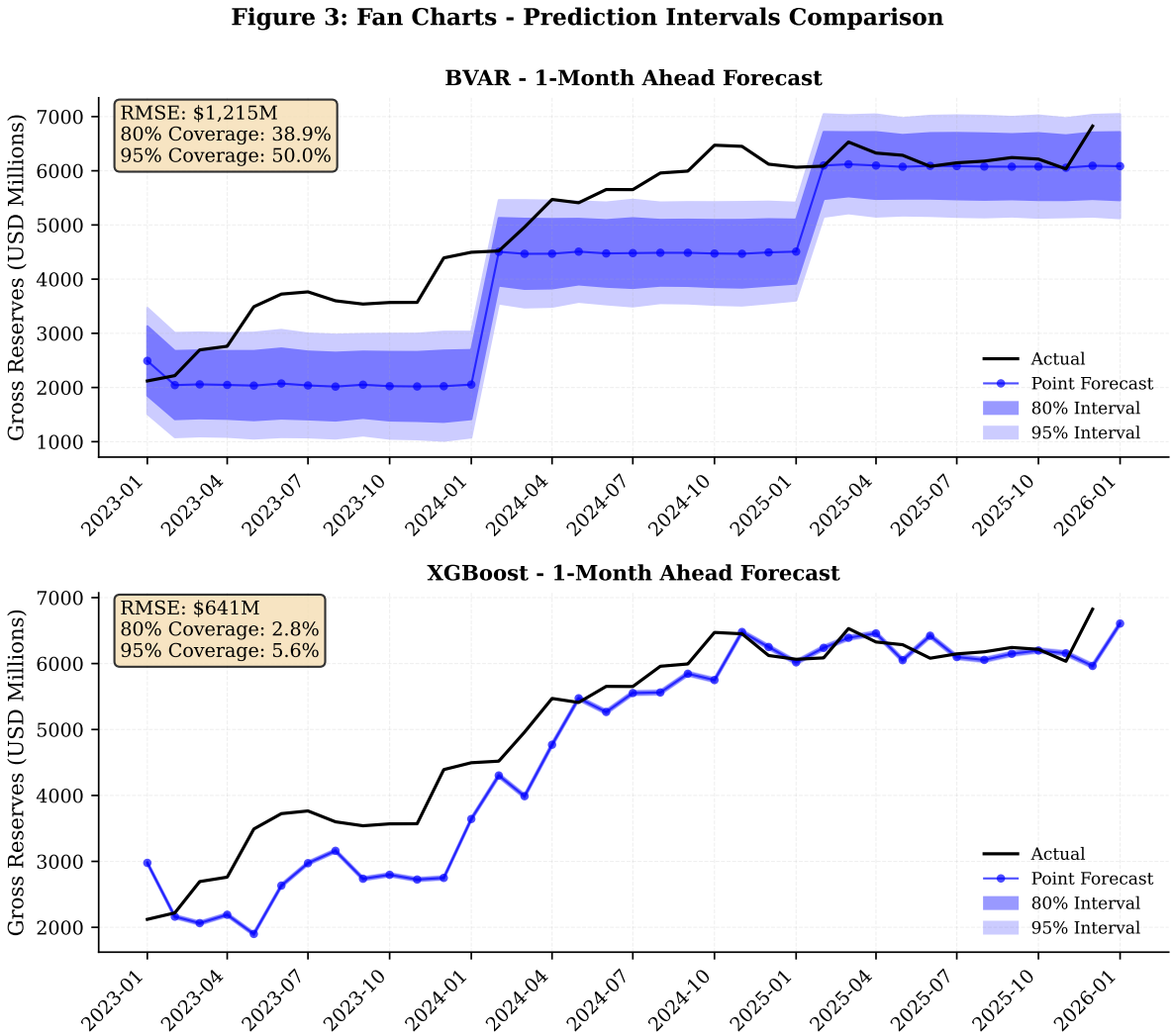

Figure 4: Prediction interval fan charts for BVAR (left) and XGB-Quantile (right). XGB-Quantile produces sharper, better-calibrated intervals; BVAR intervals are wider but exhibit substantial under-coverage relative to nominal levels.

stantially underestimate forecast uncertainty—likely because parameter estimates are unstable across regimes. The XGBoost point-forecast model produces extremely narrow implicit intervals (2.8%/5.6% coverage). Good point forecasts, in short, do not guarantee good density forecasts.

The implication for policy is a two-model approach: regime-switching models for point forecasting and scenario analysis, complemented by XGB-Quantile or conformal prediction methods for uncertainty quantification.

8 Scenario Analysis and Robustness

8.1 Scenario Analysis

Table 15: MS-VARX Scenario Analysis: Reserve Projections (Parsimonious)

Scenario	End Level	Δ (USD m)	Δ (%)
Combined Upside	6837.7	12.7	0.19
Baseline	6755.7	−69.3	−1.02
Oil Price Shock	6724.9	−100.1	−1.47
Export Shock (−15%)	6718.7	−106.3	−1.56
IMF Tranche Delay	6624.5	−200.5	−2.94
LKR Depreciation 10%	6591.7	−233.3	−3.42
Combined Adverse	6509.7	−315.3	−4.62
LKR Depreciation 20%	6427.8	−397.2	−5.82

Notes: Projections based on MS-VARX model with parsimonious specification. 12-month horizon (2025:01–2025:12). Baseline from 2024:12 level of 6,825.0. Remittance and tourism shocks have zero impact under this specification as those variables are excluded; their effects propagate in BoP and fuller specifications.

Table 15 presents component-level scenario projections over a 12-month horizon. The baseline projects reserves at 6,755.7 USD million by end-2025, a modest 1.02% decline. Under a combined adverse scenario (LKR depreciation, export shock, oil price spike, IMF disbursement delay), reserves decline to 6,509.7 million (−4.62%), while a 20% currency depreciation alone produces a 5.82% decline. IMF disbursement delays prove particularly consequential (−2.94%), underscoring the sensitivity of the recovery trajectory to programme timing.

Variable set sensitivity analysis reveals that the full specification is most vulnerable to adverse shocks (9.97% decline under combined adverse in the full specification), whilst the parsimonious specification is most stable (4.62%). The BoP specification uniquely captures remittance decline impacts (−3.78%) and tourism recovery effects (+0.89%) that are invisible in the parsimonious system.

8.2 Robustness to Train/Test Split

Table 16: Split Robustness: RMSE and Rank by Train/Validation Partition (BoP, $h = 1$)

	RMSE by Split			Rank Statistics	
Model	Baseline	Early	Late	Avg. Rank	Rank SD
MS-VAR	315.2	275.9	260.6	1.00	0.00
DMA	321.7	282.9	264.5	2.33	0.58
MS-VECM	323.4	291.4	268.9	3.33	0.58
DMS	326.0	285.4	260.6	2.67	1.53
XGBoost	886.1	1115.0	623.5	5.00	0.00

Notes: Baseline split: train to 2019-12, validate to 2022-12. Early split: train to 2018-12, validate to 2021-12. Late split: train to 2020-12, validate to 2023-12. DMS ties MS-VAR in the late split (260.6). The split-robust evaluation set excludes LSTM and BoPIdentity (configuration flags 'include_lstm = false', 'include_bop = false'); the table therefore reports the top five models within that evaluated set. Remaining evaluated models (ARIMA, Naïve, BVAR, VECM) have RMSE > 1,200 in all splits.

Table 16 addresses whether the model ranking is an artefact of one particular train/validation/test partition. It is not. Under three split definitions—baseline (train to 2019-12, validate to 2022-12), early (2018-12 / 2021-12), and late (2020-12 / 2023-12)—MS-VAR holds rank 1 in every case, with RMSEs of 315.2, 275.9, and 260.6 respectively. Its average rank is 1.00; its rank standard deviation is 0.00.

DMA is the most stable runner-up (average rank 2.33, rank SD 0.58). DMS ties MS-VAR in the late split at 260.6, but is less stable overall (average rank 2.67, rank SD 1.53). The gap between the top-four cluster and XGBoost widens in the early split, where XGBoost deteriorates to 1,115.0—nearly four times the MS-VAR error.

These results support two claims. First, the gains from regime-switching architecture are genuine, not driven by a favourable partition. Second, "stability" and "accuracy" are distinct properties. Some simpler models (e.g., Naïve) are relatively stable across splits, but at substantially worse loss levels. The regime-aware models are both accurate and stable—a combination that is rare in macroeconomic forecasting comparisons.

Split choice does, however, materially affect regime *diagnostics*: the training-validation subsample converged where the full-sample estimation often did not, and estimated regime durations differ across splits. Production forecasting systems should report results across multiple configurations and flag cases where regime diagnostics are qualitatively different.

9 Conclusion

This paper provides a comprehensive multi-model forecasting comparison for Sri Lanka's foreign exchange reserves, evaluated across five variable sets and multiple forecast horizons on a dataset encompassing stable accumulation, pandemic disruption, sovereign default, and IMF-supervised recovery. The analysis goes beyond a standard model horse-race. It introduces a formal architecture-versus-information decomposition, incorporates dynamic model combination methods, and provides mechanistic evidence for why certain models succeed.

The central result is the dominance of regime-aware methods. MS-VAR achieves RMSE reductions of roughly three-quarters relative to the naïve benchmark. DMA and DMS, which dynamically reweight constituent model forecasts, perform nearly as well—and the 90% Model Confidence Set retains all four regime-aware specifications (MS-VAR, DMA, MS-VECM, DMS) while eliminating everything else. This ranking is not an artefact of a single partition: across three split definitions, MS-VAR holds rank 1 with zero rank variance, and DMA remains the most stable runner-up. The 2×2 factorial decomposition establishes that the advantage is primarily architectural: regime-switching structure accounts for the bulk of the improvement over gradient-boosted trees, with a significant interaction effect indicating that architecture and information content are complements for regime-switching models but substitutes for tree-based methods.

The information-loss analysis demonstrates that aggregating balance-of-payments flows destroys the vast majority of the available signal through cancellation of offsetting gross flows. Regime characterisation reveals highly persistent states with near-perfect classification certainty. Impulse response analysis shows strong regime asymmetry in shock propagation that is economically consistent with distinct crisis and recovery dynamics.

The Meese-Rogoff puzzle does not hold in its strong form for reserves. MS-VAR substantially outperforms the naïve benchmark even during the acute crisis segment. A weaker form persists, however: among non-regime-switching models, the random walk remains competitive during crisis periods. XGBoost dominates in the monetary and full specifications, suggesting that tree models can exploit broader nonlinear feature spaces when dimensionality is sufficient, though their heavy reliance on autoregressive reserve features limits their ability to exploit structural BoP information.

9.1 Limitations and Future Work

Several limitations circumscribe these findings. First, although seeded multi-start checks indicate local stability around the baseline MS-VAR solution, broader random initialisations can still locate inferior local optima; regime estimation in short samples therefore remains non-trivial and should always be accompanied by reported convergence diagnostics. Second, LSTM robustness remains incompletely explored. Recurrent architectures

are sensitive to sequence length, learning rate, and regularisation; more exhaustive tuning could alter the relative ranking of neural network approaches. Third, the single-country design limits generalisability. Application to a cross-country panel of emerging markets—particularly those with different external vulnerability profiles—would test whether the MS-VAR advantage is specific to Sri Lanka or reflects a more general property of regime-switching models in crisis-prone settings. Fourth, real-time data vintage effects are not explicitly modelled. The use of revised data may overstate forecast accuracy relative to what would have been achievable in real time.

Several promising avenues emerge for future research. Conformal prediction methods offer a principled approach to combining regime-switching point-forecast accuracy with distribution-free uncertainty quantification, potentially resolving the gap between MS-VAR's point forecast dominance and XGB-Quantile's superior density calibration identified in this study. Ensemble methods combining regime-switching models with machine learning via stacking could exploit the complementary strengths of both approaches—structural regime identification from MS-VAR and flexible nonlinear feature extraction from tree-based methods. Finally, extending the information-loss framework to other macroeconomic aggregates (GDP components, monetary aggregates) could test whether the severe cancellation documented here for balance-of-payments flows is a broader phenomenon in macroeconomic forecasting.

Despite these limitations, the evidence points in a clear direction. Explicitly modelling regime dynamics in disaggregated balance-of-payments systems is essential for reserve forecasting in crisis-prone emerging markets. Central banks confronted with reserve surveillance should pair regime-switching models for scenario analysis with complementary probabilistic methods for uncertainty quantification. The stakes are high. Adequate buffers can mean the difference between weathering a sudden stop and spiralling into sovereign default.

References

Aizenman, J. (2012). The internationalization of the RMB, capital market openness, and financial reforms in China. *Pacific Economic Review*, 17(3):321–338.

Aizenman, J. and Lee, J. (2007). International reserves: precautionary versus mercantilist views, theory and evidence. *Open Economies Review*, 18(2):191–214.

Athukorala, P. (2024). Sri Lanka's external sector crisis: causes and consequences. *World Development*, 177:106556.

Banbura, M., Giannone, D., and Reichlin, L. (2010). Large Bayesian vector auto regressions. *Journal of Business & Economic Statistics*, 28(1):57–75.

Brunetti, C., Mariano, R. S., Scotti, C., and Tan, A. H. (2007). Markov switching and stochastic volatility in currency crises. *Journal of Applied Econometrics*, 22(4):765–789.

Calvo, G. A. (1996). Varieties of capital flows and their implications for the economy. In *The Economic of Globalization: Policy Perspectives from the East Asia and Pacific*, pages 15–41. World Bank, Washington, DC.

Calvo, G. A. (1998). Contagion, well-behavedness and sudden stops. *NBER Working Paper No. 5854*.

Calvo, G. A., Izquierdo, A., and Loo-Kung, R. (2013). Optimal holdings of international reserves: self-insurance against sudden stops. *Journal of International Economics*, 87(2):266–286.

Carriero, A., Clark, T. E., and Marcellino, M. (2015). Realizing the gains from Bayesian dynamic model averaging. *International Journal of Forecasting*, 31(4):1009–1023.

Central Bank of Sri Lanka (2021). Annual Report 2021. Central Bank of Sri Lanka, Colombo.

Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM.

Cheung, Y. W., Chinn, M. D., and Pascual, A. G. (2005). Empirical exchange rate models of the nineties: Are any fit to survive? *Journal of International Money and Finance*, 24(7):1150–1175.

Doan, T. A., Litterman, R. B., and Sims, C. A. (1984). Forecasting and conditional projection using realistic prior distributions. *Econometric Reviews*, 3(1):1–100.

Dooley, M. P., Lizondo, J. S., and Mathieson, D. J. (1989). The currency composition of foreign exchange reserves. *IMF Staff Papers*, 36(2):385–434.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.

Frankel, J. A. and Saravelos, G. (2012). Can we predict the next financial crisis? *NBER Working Paper No. 18725*.

Ghosh, A. R., Ostry, J. D., and Qureshi, M. S. (2017). Taming the tide of capital flows: a policy guide. *Journal of International Economics*, 105:S22–S38.

Gosselin, M.-A. and Parent, N. (2005). An empirical analysis of foreign exchange reserves in emerging Asia. *Bank of Canada Working Paper No. 2005-38*.

Greenspan, A. (1999). Currency reserves and debt. *Remarks before the World Bank Conference on Recent Trends in Reserves Management*, Washington, DC.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384.

Hansen, P. R., Lunde, A., and Nason, J. M. (2011). The model confidence set. *Econometric Reviews*, 30(2):160–201.

Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8):1735–1780.

Hyndman, R. J. and Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688.

IMF (2011). Assessing reserve adequacy. *IMF Policy Paper*.

IMF (2015). Assessing reserve adequacy—review of the adequacy approach. *IMF Policy Paper*.

IMF (2022). Sri Lanka: Request for an extended arrangement under the Extended Fund Facility. *IMF Country Report No. 23/99*.

Jeanne, O. and Rancière, R. (2011). The optimal level of international reserves for emerging market countries. *Journal of International Economics*, 85(2):229–240.

Johansen, S. (1988). Statistical analysis of cointegration vectors. *Journal of Economic Dynamics and Control*, 12(2-3):231–254.

Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6):1551–1580.

Kaminsky, G. L., Lizondo, S., and Reinhart, C. M. (1998). Leading indicators of currency crises. *IMF Staff Papers*, 45(1):1–48.

Koop, G. and Korobilis, D. (2012). Forecasting inflation using dynamic model averaging. *International Economic Review*, 53(3):867–886.

Krolzig, H. M. (1997). *Markov-Switching Vector Autoregressions: Modelling, Statistical Inference, and Application to Business Cycle Analysis*. Springer, Berlin.

Litterman, R. B. (1986). Forecasting with Bayesian vector autoregressions—five years of experience. *Journal of Business & Economic Statistics*, 4(1):25–38.

Medeiros, M. C., Vasconcelos, G. F., Veiga, Á., and Zilberman, E. (2021). Forecasting inflation in a data-rich environment: the benefits of machine learning methods. *Journal of Business & Economic Statistics*, 39(1):98–119.

Meese, R. A. and Rogoff, K. (1983). Empirical exchange rate models of the seventies: Do they fit out of sample? *Journal of International Economics*, 14(1-2):3–24.

Obstfeld, M., Shambaugh, J. C., and Taylor, A. M. (2010). Financial stability and the global safeguard system. In *NBER International Seminar on Macroeconomics 2009*, pages 9–37. University of Chicago Press.

Peria, M. S. M. (2002). A regime-switching approach to speculative attacks: a focus on European currencies. *Journal of International Economics*, 57(2):467–490.

Raftery, A. E., Kárný, M., and Ettler, P. (2010). Online prediction under model uncertainty via dynamic model averaging: application to a cold rolling mill. *Technometrics*, 52(2):52–66.

Richardson, A., Mulder, T., and Vehbi, T. (2021). Nowcasting New Zealand GDP using machine learning algorithms. *International Journal of Forecasting*, 37(2):736–759.

Sehgal, S. and Sharma, C. (2008). A study of adequacy, cost and determinants of international reserves in India. *International Research Journal of Finance and Economics*, 20:7–21.

Rossi, B. (2013). Exchange rate predictability. *Journal of Economic Literature*, 51(4):1063–1119.

Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1):1–48.

Taylor, M. P., Peel, D. A., and Sarno, L. (2001). Nonlinear mean-reversion in real exchange rates: towards a solution to the purchasing power parity puzzles. *International Journal of Finance & Economics*, 6(4):299–318.

Yu, L., Wang, S., and Lai, K. K. (2007). Foreign-exchange-rate forecasting with artificial neural networks. *International Series in Operations Research & Management Science*, 107.

UNDP (2022). The Sri Lankan economic crisis: impacts and policy responses. *UNDP Sri Lanka Report*.

Weerakoon, D. and Jayasuriya, S. (2023). The political economy of Sri Lanka's debt crisis. *South Asia: Journal of South Asian Studies*, 46(1):123–145.

Wignaraja, G. (2024). Emerging Asia's manufacturing challenge: trade, technology and structural change. *Asian Development Review*, 41(1):56–89.

Wijnholds, J. O. and Kapteyn, A. (2001). Reserve adequacy in emerging market economies. *IMF Working Paper WP/01/143*.

A Estimation Details and Metric Definitions

A.1 Classical Models

ARIMA models with exogenous regressors are estimated via maximum likelihood, with lag order selected via AIC across ARIMA(p, d, q) specifications with $p, q \in \{0, 1, 2, 3\}$ and $d \in \{0, 1\}$. VAR and VECM models are estimated via full-information maximum likelihood with lag order selected by AIC (maximum 4 lags). Cointegration rank is determined by the Johansen trace and maximum eigenvalue tests.

A.2 Bayesian VAR

The BVAR Minnesota prior is centred on a random walk with hyperparameters: overall shrinkage $\lambda \in [0.1, 1.5]$, lag decay $\mu \in [0.1, 0.8]$, and own-lag variance $\phi \in [0.1, 0.95]$, selected via grid search on in-sample BIC. The posterior is simulated via Gibbs sampling with 10,000 draws (2,000 burn-in).

A.3 Machine Learning

XGBoost: 200 boosting rounds (300 for BoP), learning rate $\eta = 0.1$, tree depth 3–5, minimum child weight 1, subsample ratio 0.8–1.0, column subsample 0.8, time-series cross-validation. Features include lagged dependent variable (lags 1–12), lagged exogenous variables (lags 1–6), and rolling 3- and 6-month means and volatilities.

LSTM: single recurrent layer of 64 units, input sequence length 12, Adam optimiser (learning rate 0.001), dropout 0.2, batch size 32, maximum 100 epochs with early stopping (patience 10).

A.4 Accuracy Metrics

$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$; $\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$; $\text{MAPE} = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right|$

$\text{MASE} = \text{MAE}/\frac{1}{n-1}\sum_{t=2}^{n}|y_t - y_{t-1}|$; Policy loss: $L(e_t) = e_t^2$ if $e_t < 0$, $0.5e_t^2$ if $e_t \geq 0$.

$\text{CRPS} = \frac{1}{n}\sum_{t=1}^{n}\int_{-\infty}^{\infty}[F_t(z) - \mathbf{1}_{z \geq y_t}]^2 dz$; Log score: $\log S_t = \log p_t(y_t)$.

Diebold-Mariano statistic: $\text{DM} = \bar{d}/\sqrt{2\pi f_d(0)/n}$, asymptotically standard normal under equal expected loss.